

Robotics [IIT-Jodhpur]

Lab 6: Jacobian

- 1 Write a Python code for finding symbolic expressions of the Jacobian for an n -link serial robot. Take DH parameters as inputs to the code.

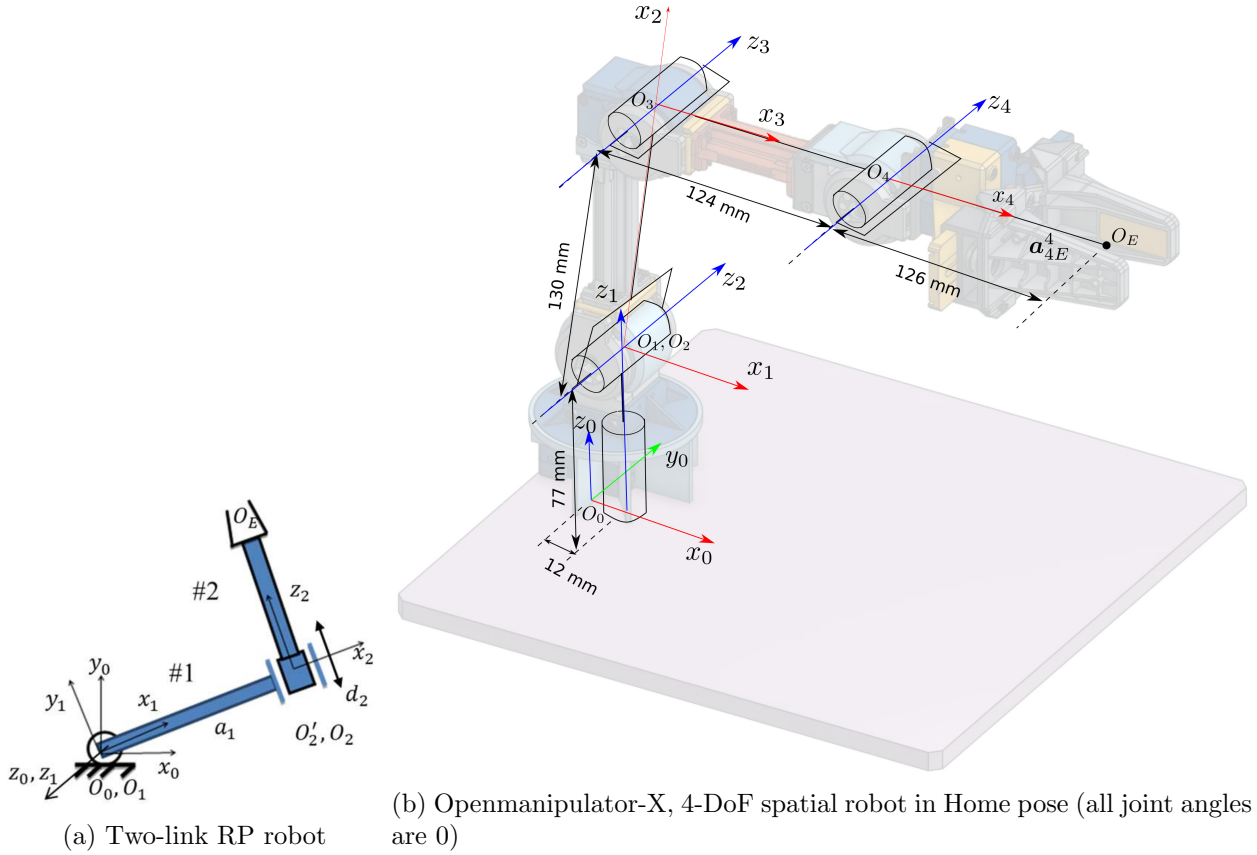
Note: For symbolic representation you can use **SymPy** Python library. It is an open-source Python library for symbolic mathematics. Unlike standard Python or numeric libraries like **NumPy** which calculate approximate floating point values, SymPy works with mathematical expressions in their exact, algebraic forms [1].

- 2 Validate code with the analytical results obtained for 2 link RP robot Figure 1a and OpenManipulator-X shown in Figure 1b for the configurations given by

(a) $\mathbf{q} = [0, 0]^\top$ for RP robot, and

(b) pose given by $\mathbf{q} = [0, -\pi/2 + \tan^{-1} \frac{0.024}{0.128}], \pi/2 - \tan^{-1} \frac{0.024}{0.128}, 0]^\top$ for Openmanipulator-X.

Interpret the results by looking at each column corresponding to a particular joint and what effect it has on the end-effector twist $\mathbf{t}_E^0$.



(a) DH Parameters for 2-link RP robot

Link	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	a_1	d_2	0

(b) DH Parameters for OpenManipulator-X for the pose shown in Fig. 1b ([2])

Link	α_{i-1} (rad)	a_{i-1} (mm)	d_i (mm)	θ_i (rad)
1	0	12	77	θ_1
2	$-\frac{\pi}{2}$	0	0	θ_2
3	0	130	0	θ_3
4	0	124	0	θ_4

Table 1: DH parameter tables

Hint

$$\mathbf{t}_E^0 = \begin{bmatrix} \dot{\mathbf{o}}_E^0 \\ \boldsymbol{\omega}_n^0 \end{bmatrix} = \begin{bmatrix} \mathbf{J}_1 & \mathbf{J}_2 & \cdots & \mathbf{J}_i & \cdots & \mathbf{J}_n \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_i \\ \vdots \\ \dot{q}_n \end{bmatrix}.$$

$$\mathbf{J}_i = \begin{bmatrix} \epsilon_i \mathbf{z}_i^0 \times (\mathbf{o}_E^0 - \mathbf{o}_i^0) + (1 - \epsilon_i) \mathbf{z}_i^0 \\ \epsilon_i \mathbf{z}_i^0 \end{bmatrix}$$

$$\epsilon_i = \begin{cases} 1, & \text{Revolute Joint,} \\ 0, & \text{Prismatic Joint,} \end{cases}$$

$$\dot{q}_i = \begin{cases} \dot{\theta}_i, & \text{Revolute Joint,} \\ \dot{d}_i, & \text{Prismatic Joint.} \end{cases}$$

Find

$$\mathbf{T}_i^0 = \begin{bmatrix} \mathbf{R}_i^0 & \mathbf{o}_i^0 \\ \mathbf{0}^T & 1 \end{bmatrix} = \mathbf{T}_1^0(q_1) \mathbf{T}_2^1(q_2) \cdots \mathbf{T}_i^{i-1}(q_i).$$

$\mathbf{z}_i^0$: third column of $\mathbf{R}_i^0$,

$\mathbf{o}_i^0$: fourth column of $\mathbf{T}_i^0$.

For the end-effector position,

$$\begin{bmatrix} \mathbf{o}_E^0 \\ 1 \end{bmatrix} = \mathbf{T}_n^0 \begin{bmatrix} \mathbf{a}_{nE}^n \\ 1 \end{bmatrix}.$$

References

- [1] SymPy Development Team, “SymPy: Python library for symbolic mathematics.” <https://www.sympy.org/en/index.html>, 2026. Accessed: September 8, 2026.
- [2] ROBOTIS, “Specification.” https://docs.robotis.com/docs/systems/openmanipulator_x/specification/, 2026. Accessed: August 19, 2026.